ANTIHUNTER
DIGI NODE

Operator's Guide: from the box to the field

Covers all four tiers: **Fully Assembled Unit** · **Complete Parts Kit** · **Populated PCB** · **Bare PCB**

Guide revision 1.0 · 14 August 2026 · Written for the current Stable and Beta firmware channels
github.com/lukeswitz/AntiHunter · Web flasher: lukeswitz.github.io/AntiHunter

Hello, operator

Welcome to AntiHunter. This node is a self-contained WiFi, Bluetooth and LoRa mesh sensor. It runs everything on the device: no cloud, no account, no server needed. It is fully open source. Read the firmware, write your own mesh commands, wire it into your own IDS. Make it yours.

This guide takes you from opening the box to a tuned node in the field, whichever tier you bought. Sections that only apply to one tier are marked. Everything else applies to every node.

READ BEFORE APPLYING POWER: THREE RULES

1. Regulated 5 V only. Power the PCB from USB, a bench supply set to 5 V, a proper DC-DC regulator, or the UPS board. Never connect 18650 cells, a battery pack, or a vehicle supply directly to the board. Unregulated input permanently damages the modules. Voltages under 4.5 V cause instability.

2. Never power the LoRa radio without its antenna. Transmitting into an open port can destroy the radio's PA. Fit the LoRa antenna, or the U.FL pigtail plus antenna, before the node is energised.

3. One USB cable at a time. The ESP32 and the Meshtastic radio share the PCB's power rails. With both USB ports plugged in, one board back-feeds the other. Plug in one board, do your work, unplug it, then plug in the other.

What you are holding

Tier	What is in the box	What you still have to do
1 · Fully Assembled Unit	Built, soldered, tested and ready to deploy. 8 GB SD card fitted. Active GPS helix antenna. SMA antennas. Meshtastic flashed on the radio with the serial link configured.	Fit two 18650 cells, screw on the antennas (§2), flash AntiHunter (§5), set the radio region and channel (§7).
2 · Complete Parts Kit	Every part on the bill of materials, unassembled, as it comes from the supplier. Passive GPS antenna. Nothing is soldered and nothing is flashed.	Solder and assemble it (§3), flash Meshtastic on the radio (§7.1), then flash AntiHunter (§5) and configure.
3 · Populated PCB	A fully assembled PCB. Core modules soldered, bench-tested, 8 GB SD card fitted. Factory U.FL antennas. Meshtastic flashed on the radio with the serial link configured.	Attach the antennas (§2), supply power over USB or 18650s, build the housing if you want one, flash AntiHunter (§5), set the radio region and channel (§7).
4 · Bare PCB	One 82 mm board, unpopulated.	Source the BOM, solder the modules, flash Meshtastic on the radio, fit an SD card, then follow this guide from §3.

TIERS 1 AND 3: THE RADIO IS ALREADY DONE

On the Fully Assembled Unit and the Populated PCB, the Meshtastic radio ships flashed with the latest stable firmware and the serial link to the node already configured: TEXTMSG mode, 115200 baud, on the correct pins for the board. You do *not* need to redo that. You *do* still need to set your LoRa region and create your own encrypted channel. See §7.

On the Complete Parts Kit and the Bare PCB you flash the radio yourself first. §7.1 has the settings. The ESP32 ships empty on *every* tier, so you always flash AntiHunter yourself and know exactly what is on it.

Contents

1 Unboxing and inventory

Open the box on a clean, static-safe surface. Check what arrived against your tier before you power anything.

1.1 Tier checklists

TIER 1: FULLY ASSEMBLED UNIT

- ☐ Node built, soldered and bench-tested
- ☐ 8 GB SD card fitted
- ☐ Active GPS helix antenna
- ☐ 2.4 GHz WiFi/BLE SMA antenna
- ☐ LoRa SMA antenna for your region
- ☐ Meshtastic flashed on the radio, serial link configured

TIER 2: COMPLETE PARTS KIT

Every part needed to build a finished node. Nothing is soldered and nothing is flashed. The two 18650 cells are the only thing you supply.

- ☐ Bare 82 mm PCB
- ☐ XIAO ESP32-S3
- ☐ Heltec V3.2 LoRa radio (or T114)
- ☐ ATGM336H GPS module
- ☐ DS3231 RTC module
- ☐ SW-420 vibration sensor
- ☐ Micro SD reader module
- ☐ 8 GB SD card
- ☐ Pin headers for every module
- ☐ 3D-printed enclosure: main PCB housing, battery housing, front cover, back cover
- ☐ TPU seals for the housing, GPS and USB-C port
- ☐ 3× U.FL-to-SMA pigtail cables

- ☐ SMA bulkheads
- ☐ 2.4 GHz WiFi/BLE antenna, 6 dBi
- ☐ LoRa antenna for your region
- ☐ Passive ceramic GPS antenna, the one supplied with the GPS module
- ☐ 30 mm 5 V fan with JST lead
- ☐ KSD9700 normally-open thermal switch, 40 °C
- ☐ 3-pin mini on/off switch with rubber cover
- ☐ Waterproof Type-C female chassis jack, 2-pin 22 AWG leads, panel nut and dust cap
- ☐ Type-C 15 W 3 A 5 V UPS charger board, dual 18650
- ☐ 6× JST 2.54 2-pin terminals
- ☐ 3× JST power male cables for the switch, thermal switch and power board
- ☐ M3 heat-set inserts, plus 4× M2 for the power board
- ☐ M3×4-6mm screws for the PCB and enclosure lids, M2×4-6mm for the power board
- ☐ 2× M2.5 screws, supplied with the fan
- ☐ 2× 15 mm brass standoffs
- ☐ ¼"-20 tripod insert

TIER 3: POPULATED PCB

A fully assembled PCB, bench-tested, on its factory antennas. No enclosure and no power source.

- ☐ PCB with XIAO ESP32-S3, Heltec V3.2 LoRa radio (or T114), ATGM336H GPS, DS3231 RTC, SW-420 vibration sensor and SD reader soldered
- ☐ 8 GB SD card fitted
- ☐ Factory U.FL antennas: flat 2.4 GHz film, coiled LoRa whip, ceramic GPS patch
- ☐ Meshtastic flashed on the radio, serial link configured

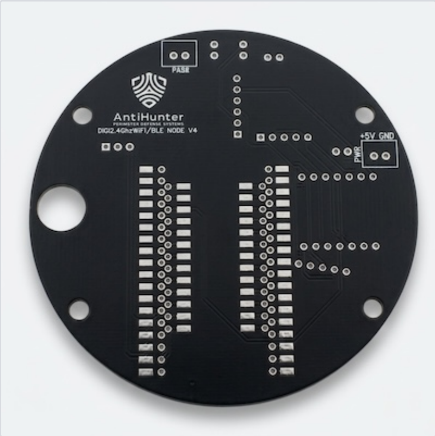
TIER 4: BARE PCB

- ☐ One 82 mm AntiHunter DIGI NODE PCB, unpopulated

Everything else comes from the bill of materials in Appendix B, including the Meshtastic radio, which you flash yourself.

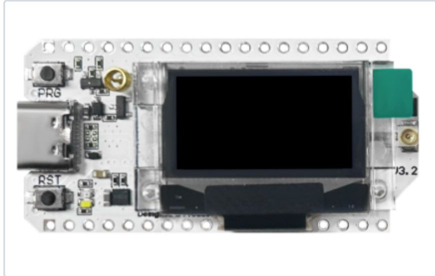
1.2 What you supply

- **Power.** Tiers 1 and 2 take two 18650 cells in the UPS board. Tier 3 runs from USB or from your own regulated 5 V supply. Cells are never supplied on any tier.
- **18650 cells.** Brand new, same brand, same capacity, never mixed old with new. Sourcing, charging, storage and disposal are your responsibility.
- **A micro SD card**, only on the Bare PCB tier. Every other tier ships with an 8 GB card. A replacement must be FAT32, and 32 GB or larger is not recommended.
- **A USB-C data cable.** Charge-only cables will not flash the board.
- **A computer running Chrome or Edge** for the web flasher, or a terminal for the CLI installer.
- **Soldering tools** on tiers 2 and 4: temperature-controlled iron, fine solder, flux, tweezers, small Phillips and hex drivers, heat-resistant surface.

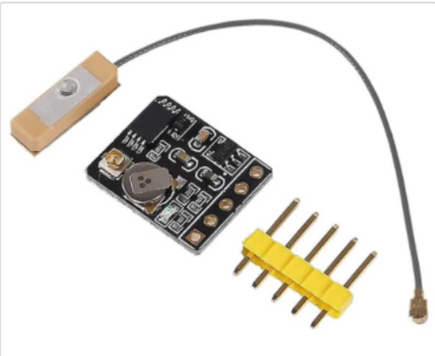


The bare 82 mm board.

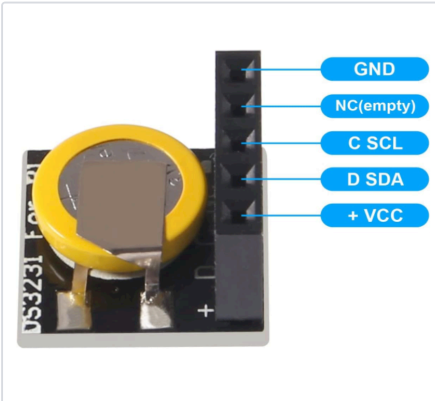
XIAO ESP32-S3.



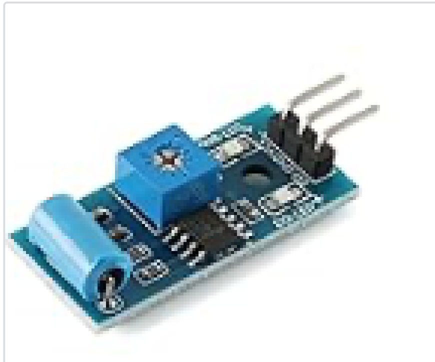
Heltec V3.2 LoRa radio.



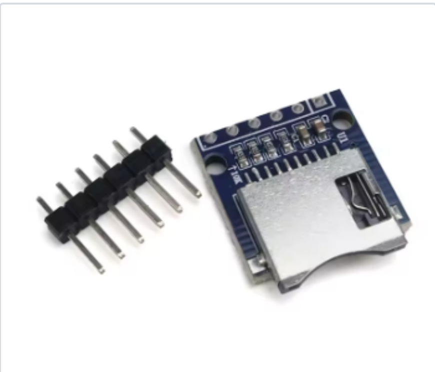
ATGM336H GPS module.



DS3231 real-time clock.



SW-420 vibration sensor. Note the trimpot.



Micro SD reader module.

U.FL-to-SMA pigtail.

5 V UPS charger board.

1.3 Power

The node draws roughly 160 mA idle and 600 mA while actively scanning. Inside the enclosure the UPS board provides the regulated 5 V and the cells are never connected to the PCB directly. Outside the enclosure, use a USB-C adapter, a bench supply set to 5.0 V on the 5 V input, or a proper DC-DC regulator.

> **POWER RULES**
>
> Do not connect batteries directly. Do not exceed 5 V. Do not use unregulated sources. A fuse in the battery line is recommended, a 2 A fast-blow in series with the switch wiring, but one is not supplied with any tier.

1.4 Thermal

The ESP32 runs warm under load and the GPS module gets noticeably hot while it searches for satellites. Both are normal. Neither the fan nor the thermal switch is required for the node to work.

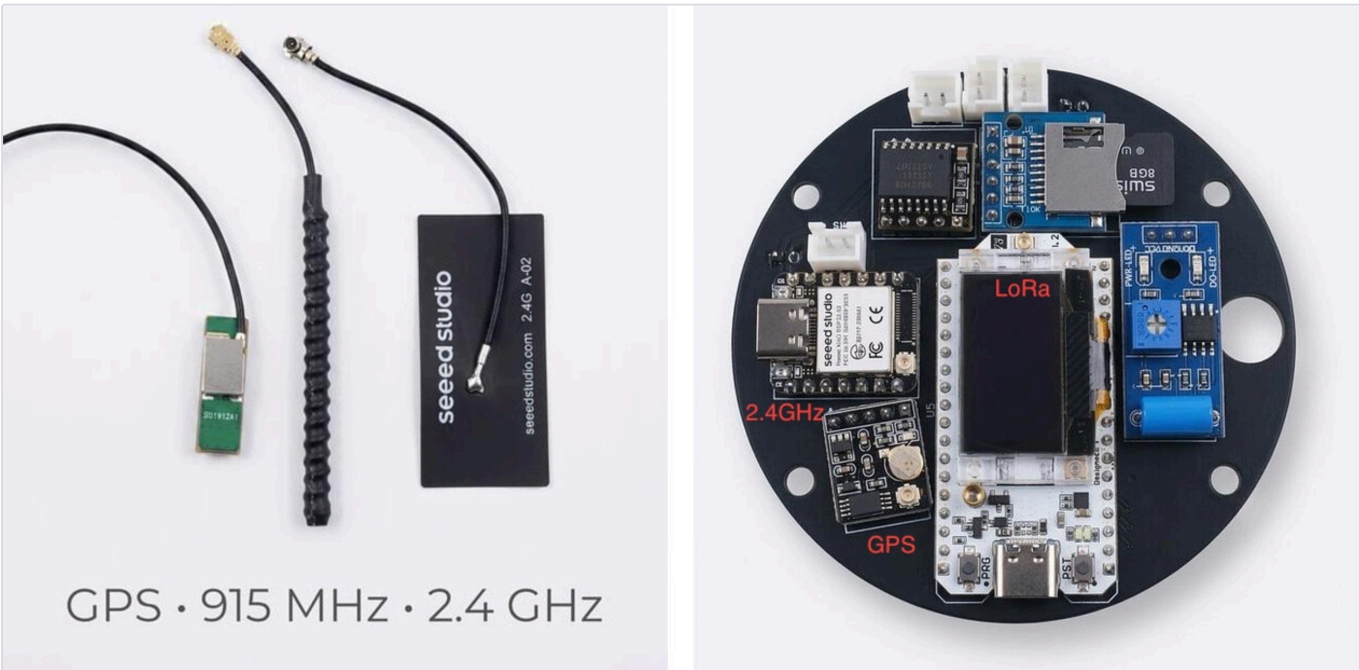
- The 30 mm fan plugs into the FAN JST output.
- The sticker side is not always the exhaust side. Run the fan for a second and feel which way it blows before you screw it down. The Fully Assembled Unit ships with the fan set to exhaust.
- The KSD9700 normally-open thermal switch, 40 °C, wires to the THERMO JST and switches the fan on when the enclosure heats up.
- To run the fan continuously instead, jump the THERMO input or supply 5 V straight to the fan port.

2 Antennas

More nodes are damaged fitting antennas than in any other step. The U.FL connectors are small, fragile and unforgiving of a crooked press. Take your time here.

2.1 Which antenna goes where

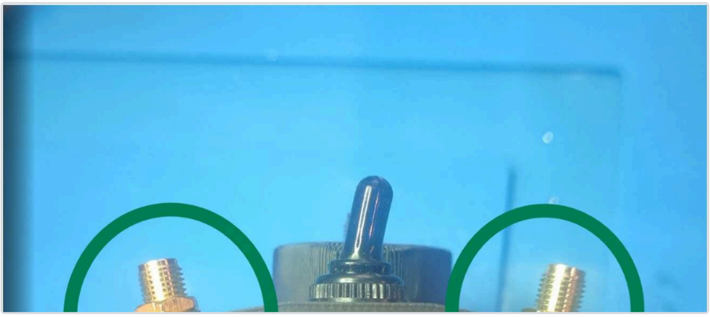
Antenna	U.FL set (tiers 2 and 3)	SMA set (tiers 1 and 2)	Connects to
2.4 GHz WiFi / BLE	Flat black film antenna, printed `2.4G`	Shorter of the two rubber ducks, 6 dBi	The U.FL socket on the XIAO ESP32-S3, or its SMA bulkhead
LoRa (region specific)	Coiled black whip	Longer duck, marked 868 MHz (EU), 915 MHz (US) or 923 MHz (Asia)	The U.FL socket on the Heltec / T114 radio, or its SMA bulkhead
GPS	Passive ceramic patch, supplied with the GPS module	Active helix, Fully Assembled Unit only	The U.FL socket on the GPS module, or the SMA bulkhead in the battery housing



The factory U.FL antenna set beside the sockets each one belongs to: ceramic GPS patch, coiled LoRa whip (915 MHz shown, yours matches your region), flat 2.4 GHz film antenna. The GPS module carries the GPS socket, the Heltec carries LoRa, the XIAO carries 2.4 GHz WiFi and BLE.

PICK AN ORIENTATION AND KEEP IT

On an enclosed build, decide which bulkhead carries LoRa and which carries WiFi, then wire every node you build the same way. It does not matter which you choose. It matters that a spare antenna always fits the port you expect in the field.



The two SMA bulkheads with the power switch centred between them.

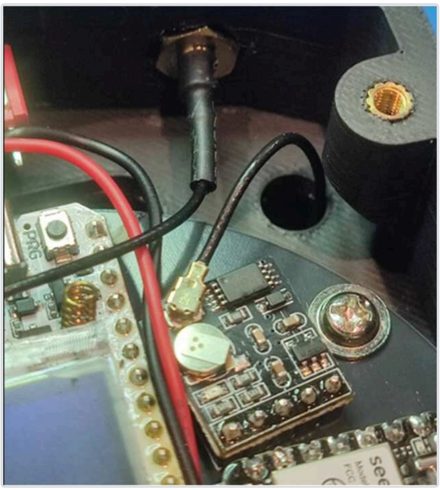
MATCH YOUR ANTENNAS ACROSS NODES

When you build more than one node, use the same antenna type and the same cable length on all of them. Triangulation and any other multi-node distance estimate assumes the nodes hear the world the same way. Mismatched antennas skew the result.

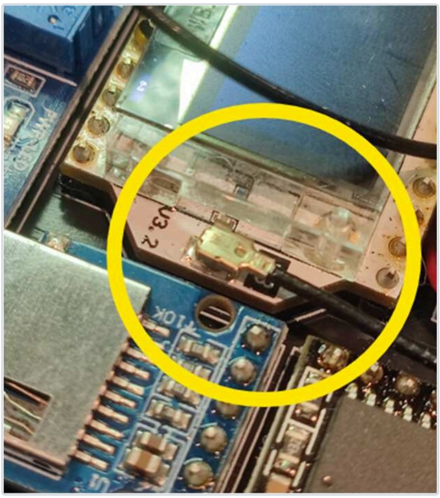
2.2 Fitting a U.FL / IPX pigtail (tiers 2 and 3)

The pigtail's tiny snap connector mates with the socket on each module. It seats with an audible, tactile click.

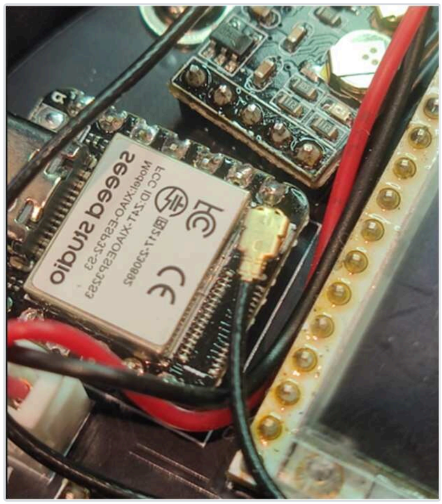
1. Identify each pigtail and the socket it belongs to. Route the cable before you connect it, not after.
2. Hold the connector **parallel to the board** and align its rim concentrically over the socket. Look at it from the side: the two rims must be perfectly stacked.
3. Press **straight down** with a fingertip or the flat of a plastic tool until it clicks.
4. Tug the cable gently to confirm it is home. If it lifts off, it never seated.



GPS module, connector facing up.



LoRa radio, socket circled.



XIAO ESP32-S3, WiFi/BLE socket.

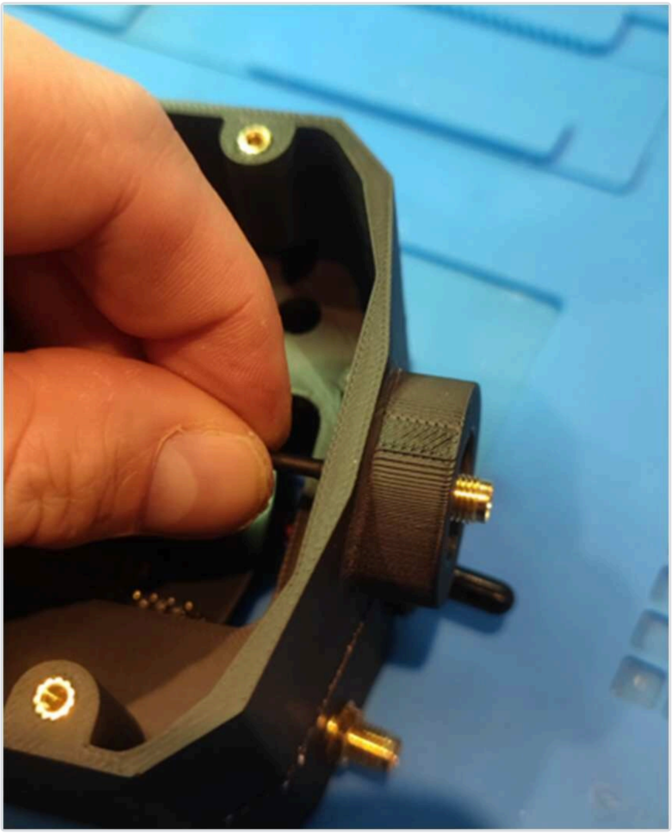
U.FL: DO NOT

- Do not press at an angle.
- Do not force a connector that will not click.
- Do not pull the cable to remove one. Lift the connector body straight up, or use a U.FL removal tool.
- Do not reseat one repeatedly. The contacts wear out.

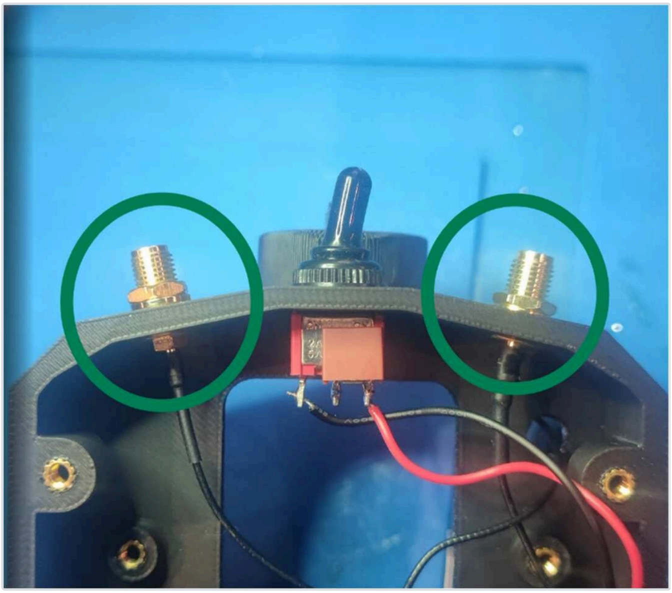
- **Misalignment damage is permanent.** A crushed socket means a new module.

2.3 Fitting SMA antennas (tiers 1 and 2)

1. Check the bulkhead is tight in the shell. The nut and washer go on the outside, and the bulkhead should not rotate when you turn an antenna onto it.
2. Start the thread by hand, straight on. Never cross-thread it.
3. Turn until finger-tight, then no more than an eighth of a turn. Do not use pliers.
4. Fold the duck upright once fitted. Antennas work best vertical.



GPS SMA female fitted into the battery housing, secured from outside with washer and nut.



WiFi SMA (female, left) and LoRa SMA (male, right) in the main housing.

WET CONDITIONS

A smear of silicone grease on the SMA threads and on the TPU seals keeps water out and makes the antennas easier to remove later.

3 Assembly (tiers 2 and 4)

Tiers 1 and 3 arrive soldered. Skip to §4.

BUILD FROM THE ASSEMBLY MANUAL, NOT FROM HERE

The Complete Parts Kit and the Bare PCB are built with the illustrated assembly manual, which covers soldering, connectors, the enclosure, the SD card and the power wiring in photographed steps. This guide does not repeat it.

> `hw/Prototype_STL_Files/Antihunter-DIGINODE-AssemblyManual.pdf`
> `github.com/lukeswitz/AntiHunter`, in the `hw` folder
>
> Work through it end to end, then come back here at §4 to power the node up for the first time.

4 First power-on

4.1 What a healthy boot looks like

Connect a USB-C cable to the **ESP32** only, remembering one USB at a time, and open a serial monitor at 115200 baud. On boot the firmware prints its banner and version, then the results of its hardware checks: SD card, GPS, RTC, vibration sensor. Any module that failed to initialise says so by name.

If you flashed the **Full** firmware, a WiFi access point named `Antihunter` also appears within a few seconds.

4.2 Confirming the hardware

Every subsystem reports into the diagnostics text, visible in the web UI's System tab, over serial, and in the reply to a mesh `STATUS` command. Check for:

- **SD card.** Mounted and writable. If it is missing, the node still scans but nothing is logged.
- **GPS.** Reports a fix once it has one. Give it a clear view of the sky; indoors it may never lock. Detections carry coordinates only when the fix is valid.
- **RTC.** Supplies the timestamps on every log line and export.
- **Vibration sensor.** Shows `Standby`, or `Active` with the time of the last movement.

4.3 The clock

Nothing to do. The DS3231 is set when you flash the firmware, and GPS disciplines it again on every fix. Its coin cell keeps it running across power cycles. All timestamps in this system are UTC.

> **ORDER OF OPERATIONS**
>
> Flash the firmware first (§5), then set the clock and configure the radio. A fresh flash with "Erase Device" ticked clears saved settings, so do it before you spend time configuring.

5 Flashing the firmware

The ESP32 ships empty on every tier. You choose the build and put it on yourself.

5.1 Full or Headless?

Same firmware, same detectors, same mesh commands, same scan engine. The difference is whether the node hosts a WiFi access point.

	Full	Headless
Control	Web UI and HTTP API over its own AP, plus serial and mesh	Serial and mesh only
RF footprint	The AP beacons continuously, so the node is discoverable as a WiFi network	Never beacons

Results	Rendered as cards in the web UI's Results tab, and served plain over the API like every other endpoint. Written to the SD card as well	Written to `/last_results.txt` on the SD card
Logging	Identical. Both builds write results, the probe database, vibration events, sentinel incidents and scan logs to the SD card	
Device database	`/devicedb.jsonl`, which is what the Data Explorer browses	Not written
Best for	Bench work, first setup, anything where you want the UI	Covert and long-term field deployments

Start with Full while you learn the device. Reflash to Headless before a quiet deployment. It takes two minutes and your settings live on the SD card.

5.2 Release channels

Channel	Source	Use it when
Stable	`main` branch	Default. Field deployments.
Beta	`beta` branch	New features first, with the risk that implies.
Experimental	ESP32-C5 builds (2.4 + 5 GHz) and RadarNode	Only on C5 hardware. The flasher asks you to acknowledge these are test builds.

5.3 Method A: web flasher (recommended)

1. Open **lukeswitz.github.io/AntiHunter** in Chrome or Edge on a desktop. Other browsers cannot talk to serial ports.
2. Choose **Full** or **Headless**, and a release channel.
3. Plug the ESP32 in with a data cable, nothing else connected to the board.
4. Click **Connect & Flash** and pick the port.
5. Optionally set your device parameters before flashing. The **Sentinel & Detectors** panel configures the whole detection engine: start-on-boot, radio mode, every detector toggle, mesh flags, thresholds. Anything left on *Default* keeps the firmware's own setting.

> **ERASE DEVICE**
>
> Tick it when upgrading from firmware older than v0.9.2, or when you want to clear saved settings. Otherwise leave it off. Settings persist in NVS and are mirrored to the SD card, and self-heal if they are corrupted.

5.4 Method B: command line

```
curl -fsSL -o flashAntihunter.sh \
  https://raw.githubusercontent.com/lukeswitz/AntiHunter/main/Dist/flashAntihunter.sh
chmod +x flashAntihunter.sh
./flashAntihunter.sh
```

The script asks for a release channel, then Full or Headless. Flags: `-c` configure device parameters during the flash, `-e` erase flash first, `-l` list available firmware.

5.5 Method C: build from source

```
git clone https://github.com/lukeswitz/AntiHunter.git
cd AntiHunter
pio device list                               # find the port
pio run -e AntiHunter-full -t upload          # web UI build
pio run -e AntiHunter-headless -t upload      # serial + mesh build
pio device monitor -e AntiHunter-full         # watch it boot
pio run -e AntiHunter-full -t erase -t upload # clean flash
```

Environments: `AntiHunter-full` and `AntiHunter-headless` for the ESP32-S3; `AntiHunter-c5-full` and `AntiHunter-c5-headless` for the ESP32-C5 on the `feat/c5` branch; `RadarNode-c5` for the 24 GHz radar node.

5.6 Verifying the flash

- The serial monitor prints the banner and firmware version at boot.
- Full build: the `Antihunter` access point appears.
- Connect to it, password `antihunt3r123`, and open **http://192.168.4.1**. Note `http`, not https.

> **DO THIS BEFORE ANYTHING ELSE**
>
> The default AP name and password are published in the documentation. Until you change them in **RF Settings**, treat the node as an open door. §8.1.

5.7 Board variants

The XIAO ESP32-C5 is a drop-in replacement for the S3 on the same board, with the same pads and peripherals, and it adds 5 GHz scanning. Those builds are in testing and live under the flasher's Experimental channel. The RadarNode variant pairs a 24 GHz radar with the same mesh: it detects a moving target, then sweeps WiFi and BLE to record which devices were present at that moment.

6 Staying up to date

6.1 Where updates come from

Firmware is built and published by CI when a release is tagged. The web flasher always serves the current build for the channel you pick, so re-running the flasher is the update. Nothing on the node phones home or updates itself.

6.2 Getting notified

1. Open **github.com/lukeswitz/AntiHunter** and sign in.
2. Click **Watch** → **Custom** → tick **Releases** → Apply. You now get an email or notification for every release, without the noise of every commit.
3. Read the release notes before flashing: `docs/release-notes/notes-stable.md` and `notes-beta.md` in the repository, and on each release page.

Discussions and issues on the same repository are where behaviour changes, hardware alternatives and field reports get talked through.

6.3 Updating a deployed node

1. Bring it in, or reach it over USB in place.

2. Re-run the web flasher or the CLI script on the same channel. Leave **Erase Device** unticked to keep your settings.
3. Reconnect and check the version, then confirm your targets, node ID and detector settings survived.

Settings live in NVS and are mirrored to the SD card, so they normally survive an update. Watchlists, results and databases live on the SD card and are untouched by flashing.

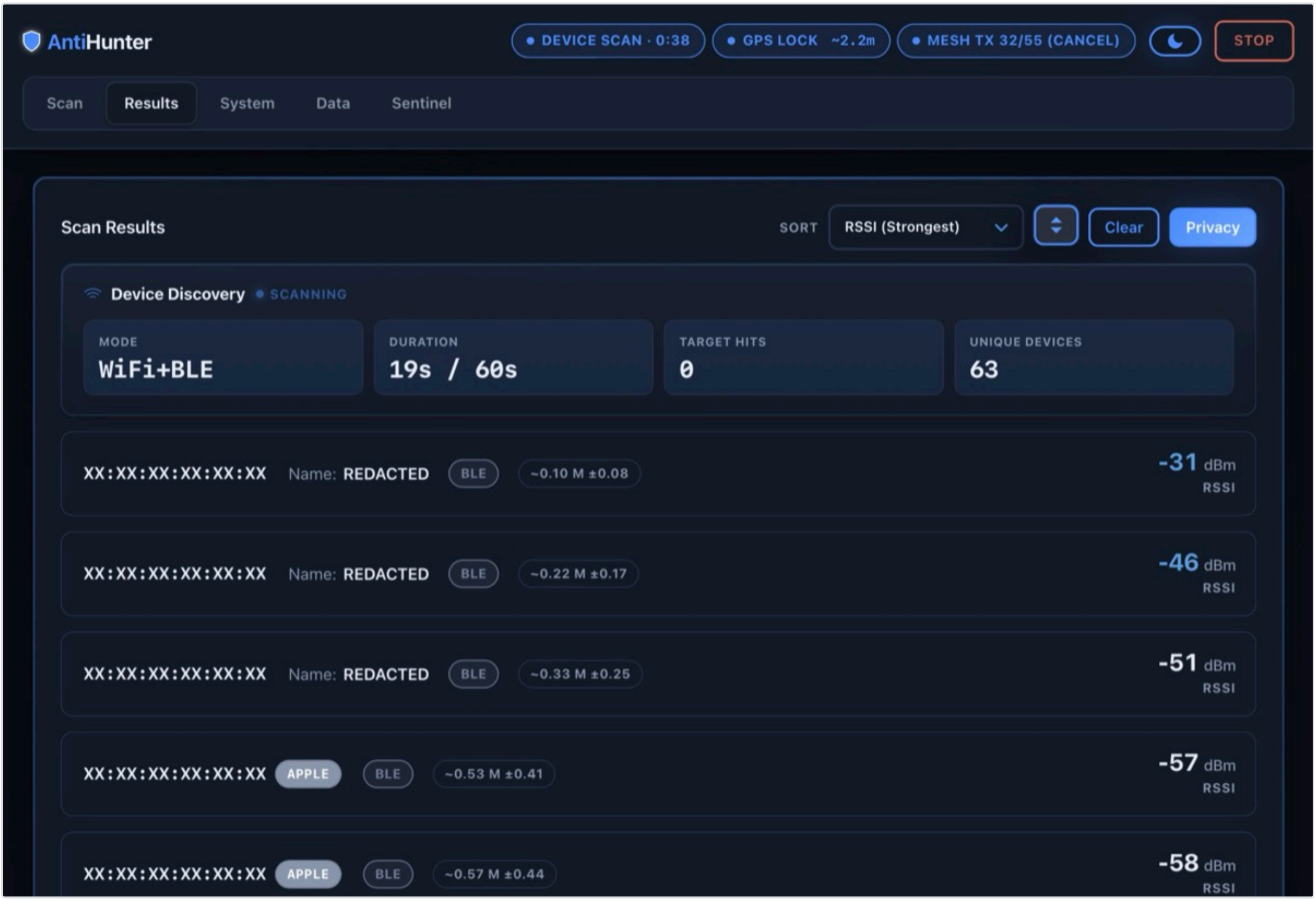
6.4 Rolling back

Flash the other channel, or check out an earlier tag and build from source with PlatformIO. If a downgrade behaves strangely, erase and flash clean, then reconfigure.

> **FLEET DISCIPLINE**
>
> Run one firmware version across a mesh. Mixed versions can differ in the mesh message formats they emit and what your Command Center is able to parse.

7 Mesh radio setup

The LoRa mesh gives you long-range control and alerting with no infrastructure. It is entirely optional, since a single node works standalone, but it is what turns several nodes into a system.



A device scan sending its results over the mesh. The header badge tracks the transmit queue draining at the LoRa airtime cap, 32 of 55 rows sent, and Privacy Mode has blanked the addresses.

7.1 Flashing and linking the radio

Tiers 1 and 3: already done. On the Fully Assembled Unit and the Populated PCB the radio arrives on the latest stable Meshtastic with the serial link to the node set: **TEXTMSG** mode, **115200** baud, on the right pins for the board (19 RX / 20 TX on Heltec V3, 10 RX / 9 TX on T114). The screen is set to blank after 1 s and the status LED is off. Verify it if you like; do not redo it.

Tiers 2 and 4: you flash it. The Complete Parts Kit and the Bare PCB include no flashed firmware at all. Connect the Heltec V3.2 (preferred) or T114 on its own USB and flash the latest stable Meshtastic from **flasher.meshtastic.org**. Then set the serial module, either in the Meshtastic app or web client, or with the script in §7.2:

Setting	Value
Serial module	Enabled
Mode	`TEXTMSG`
Baud	`115200`
Pins, Heltec V3	19 RX / 20 TX
Pins, T114	10 RX / 9 TX

Without this the node and the radio cannot talk, and no mesh command will ever reach the node.

> **THREE THINGS EVERY RADIO NEEDS, WHATEVER TIER YOU BOUGHT**
>
> - **LoRa region.** Ships `UNSET`. The radio receives but will not transmit until you set it (§7.2).
> - **Bluetooth pairing PIN.** Every unit ships with the same pin. Change it, or turn BLE off.
> - **Channel.** Ships on the public default. Make your own encrypted channel primary and turn the public one off (§7.3).

7.2 What you must set: region

The LoRa region is not set at the factory, and an `UNSET` region makes the radio receive-only: it hears the mesh but never transmits. Set the region for the country you are in. It determines the frequency band and the legal power and duty-cycle limits.

Plug the **radio** in on its own USB, not the ESP32, and run:

```
pip3 install 'meshtastic[cli]' pyserial
python3 scripts/meshtastic_config.py
```

With no arguments it prints the current settings and opens an interactive menu. Non-interactive equivalent:

```
python3 scripts/meshtastic_config.py --serial on --board heltec-v3 \
    --screen off --led off --ble off --region US
```

Option	What it does
`--port`	Serial port, auto-detected if omitted
`--board heltec-v3 | t114`	Sets the serial-module pins for your board
`--region`	LoRa region, e.g. US. `UNSET` means receive only
`--serial on | off`	The node link: TEXTMSG at 115200 on the board's pins
`--screen on | off | SECS`	`off` blanks after 1 s, or give a timeout in seconds

`--led on	off`	Status LED heartbeat
`--ble on	off`	Bluetooth radio
`--pin NNNNNN	none	random`	BLE pairing: fixed 6-digit PIN, none, or random

The same settings can be applied from the Meshtastic phone app or web client if you prefer.

7.3 Your own encrypted channel

Meshtastic ships with a public default channel whose key is published. Anything you send on it, commands, detections and coordinates alike, is readable by anyone in radio range running stock firmware. Before you deploy:

1. In the Meshtastic app, create a **new channel** with a generated key and give it a name of your own.
2. Make it the **primary** channel.
3. **Turn the public default channel off.**
4. Share the channel to your other nodes and to your phone by QR code or channel URL. Every node in your mesh must carry the same channel.

> **THE FACTORY PRESET IS NOT PRIVATE**
>
> A node left on the default channel broadcasts its detections, its position and its command traffic to anyone listening. Changing the channel is the single most important mesh setting you will touch.

7.4 Quiet-ops settings

- **Screen off** and **LED off** removes the visual signature and saves power. Physically removing the Heltec screen saves more.
- **BLE off** on the radio removes its Bluetooth signature. Set a pairing PIN first if you intend to turn it back on later.
- **Headless firmware** on the node gives no WiFi AP at all.

7.5 Addressing and node IDs

- Node IDs are 2 to 5 alphanumeric characters, A-Z and 0-9, no spaces. Give every node a distinct one.
- `@ALL COMMAND` broadcasts to every node. `@AH01 COMMAND` targets one.
- Outbound messages are rate-limited to protect airtime: roughly 167 bytes per second sustained, with device rows packed into frames of up to 230 bytes.

7.6 Reaching the mesh from a phone, TAK or MQTT

Commands and detections are ordinary Meshtastic text messages, so anything that talks to the mesh talks to your nodes:

- **Phone.** Pair the radio to the Meshtastic app over Bluetooth or WiFi and send `@node COMMAND` messages from anywhere in mesh range. No AP, no server.
- **TAK / ATAK.** The Meshtastic ATAK plugin and TAK server integration bridge the whole mesh, node traffic included.
- **MQTT.** A Meshtastic MQTT gateway node forwards mesh traffic to a broker for logging, Home Assistant or Node-RED.

These are Meshtastic features. Nothing AntiHunter-specific is needed on the receiving end.

8 First configuration

Ten minutes on the bench before the node ever leaves the building.

8.1 Change the credentials

- **AP SSID and password.** The defaults `Antihunter` and `antihunt3r123` are published. Change both under **RF Settings**.
- **Erase PSK.** The pre-shared key that authorises remote erase and factory reset. Set it if you plan to use secure data destruction (§10.4). Mesh: `@AH01 CONFIG_ERASE_PSK:yourkey`.
- **Meshtastic BLE pairing PIN.** Set a fixed PIN or turn BLE off entirely.

8.2 Identity

- **Node ID.** Set it in the web UI or with `@AH01 CONFIG_NODEID:AH02`. It prefixes every mesh message the node sends, so make it meaningful.
- **Clock.** See §4.3. Everything is timestamped UTC.

8.3 Scan presets

Presets set the whole timing profile in one click. Start on Balanced.

Preset	WiFi chan time	WiFi scan int	BLE scan int	BLE scan dur	RSSI	Use case
Relaxed	300 ms	5000 ms	6000 ms	3000 ms	–80 dBm	Low power
Balanced	160 ms	3000 ms	4000 ms	2000 ms	–95 dBm	General use (default)
Aggressive	110 ms	1500 ms	2000 ms	1000 ms	–100 dBm	Fast detection, high coverage
Custom	User-defined					Fine-tuned

What each parameter does

- **WiFi channel time**, 50 to 300 ms. Dwell per channel. It must clear the ~100 ms beacon interval to catch every AP on a channel. Shorter covers the band faster but risks missing beacons.
- **WiFi scan interval**, 1000 to 10000 ms. How often the all-channel AP discovery scan runs. Between scans the node captures frames passively while hopping.
- **BLE scan interval**, 1000 to 10000 ms. Time between BLE cycles.
- **BLE scan duration**, 1000 to 5000 ms. Active BLE scanning per cycle.
- **RSSI threshold**, –100 to –10 dBm. Global signal filter. Triangulation ignores it.
- **WiFi channels.** Comma-separated, `1,6,11`, or a range, `1..14`. Default is 1 to 11.

> **ONE RADIO, TWO JOBS**
>
> WiFi and BLE share a single 2.4 GHz radio and the scan loop is single-threaded. A BLE scan holds the radio for its full duration, and WiFi capture is off-air for that time. BLE scan duration is literally the fraction of each cycle WiFi is dark. The presets split it evenly by setting BLE duration to half the BLE interval.

8.4 Field controls worth knowing now

- **Privacy Mode.** One click blanks MACs, SSIDs and GPS in the web UI for screenshots and demos. **Exports and SD data are not redacted.** Scrub those separately.
- **Allowlist.** A global list of devices to ignore across every scan mode. Put your own kit on it.

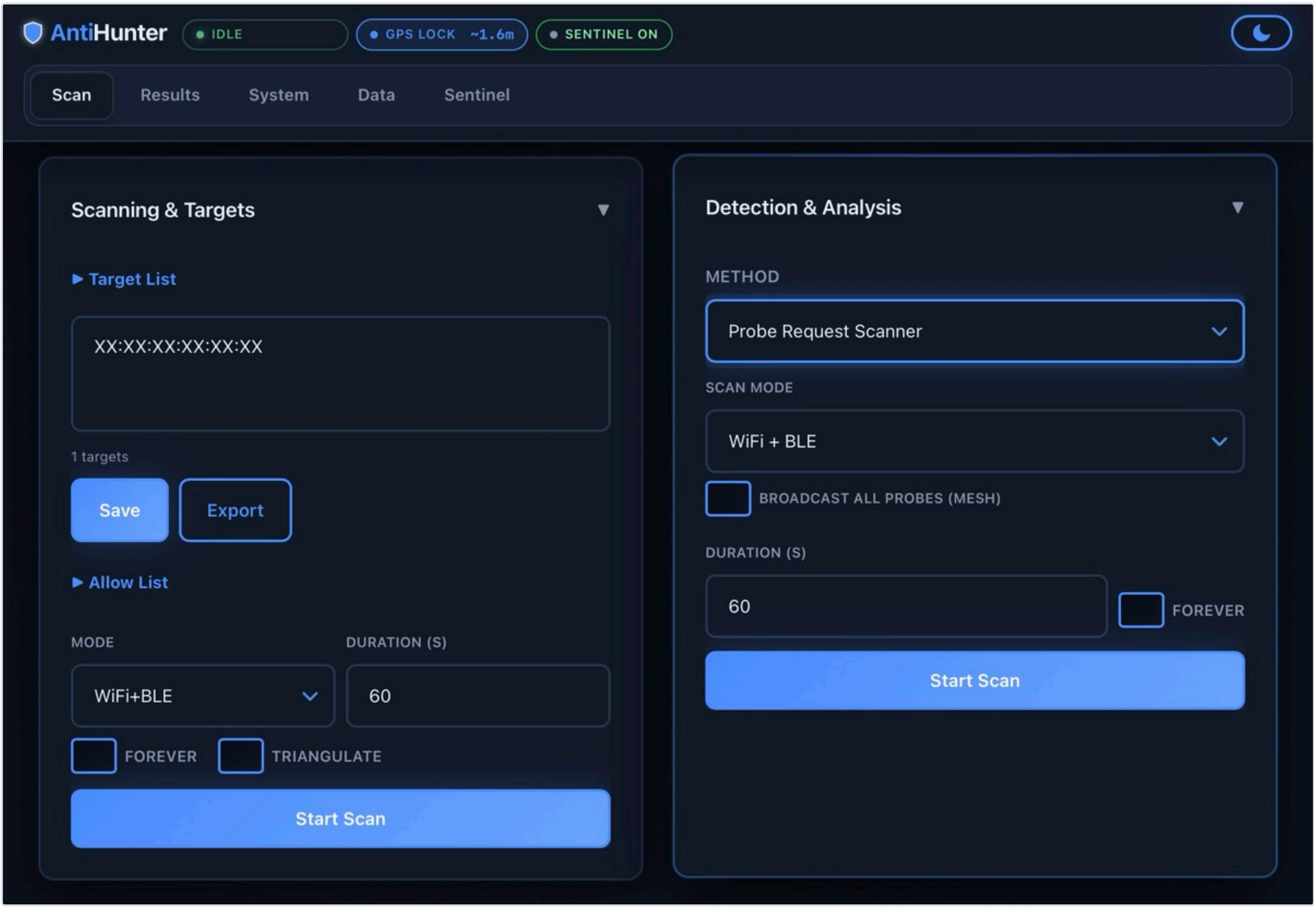
- **Battery Saver.** Stops WiFi and BLE scanning, drops the CPU to 80 MHz, enables light sleep and polls GPS once a minute. Mesh stays up. Start it with `@AH01 BATTERY_SAVER_START:10`.
- **Mesh dedup TTL.** Suppresses re-broadcasting a MAC already sent within the window, default 300 s, max 3600. Saves airtime on repeated scans of the same environment. Never applied to alerts, drone events or triangulation.

8.5 Where settings live

Configuration is written to NVS on the ESP32 and mirrored to the SD card, so it survives reboots and normal reflashes, and self-heals if a copy is corrupted. Ticking **Erase Device** at flash time wipes NVS deliberately.

9 Using the node

Every capability runs on the node itself and reports to the web UI, the mesh and the Command Center alike. One scan mode runs at a time.



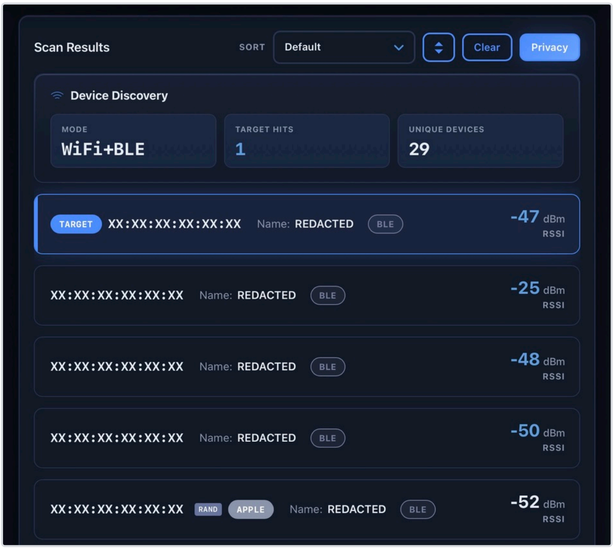
The Scan tab on the Full firmware: watchlist and scan mode on the left, detection method on the right. Every one of these runs from a mesh command too.

9.1 Target Scan

A watchlist scan. Load MAC addresses, OUI prefixes, the first three bytes of a MAC so a whole vendor, or SSIDs, and get an instant mesh alert when one is seen.

```
@ALL CONFIG_TARGETS:AA:BB:CC:DD:EE:FF|11:22:33|MyNetwork
@ALL SCAN_START:2:300:1..11     # mode 0=WiFi 1=BLE 2=both : seconds : channels
```

Alert: `NODE: Target: TYPE MAC RSSI:dBm [Name:name] [GPS=lat,lon]`



Target scan results.

9.2 Device Scanner

Captures everything nearby, WiFi and BLE, with RSSI, channel and names. This is your baseline survey tool. Run it first at any new site to see what normal looks like.

```
@ALL DEVICE_SCAN_START:2:300:+PROBE   # +PROBE also fills the probe database
```

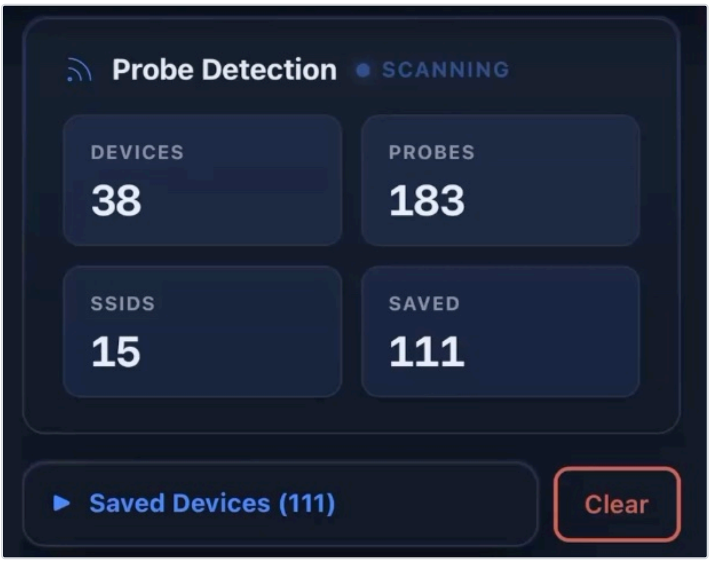
9.3 Probe Request Scanner

Passive sniffing of the probe requests devices emit while looking for networks they know. It reveals the SSIDs a device is searching for, which is often more identifying than the device itself.

```
@ALL PROBE_START:2:300:+ALL     # +ALL broadcasts every probe, not just watchlist hits
```

Hits: `NODE: PROBE_HIT MAC [Randomized|Vendor] RSSI=dBm CH=N [SSID="net" [GHOST]]`

Ghost SSID detection flags a probed SSID with no matching AP responding nearby: a network the device remembers but that is not here.



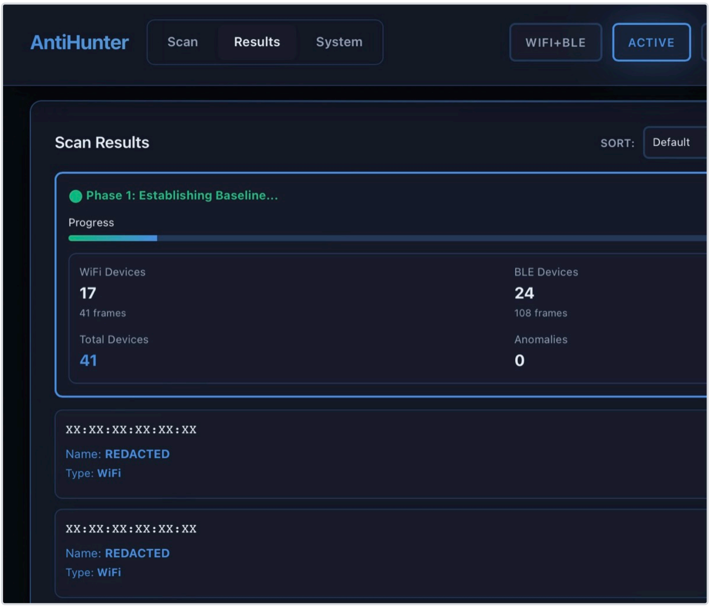
Probe detection: devices seen, probe requests captured, distinct SSIDs, and saved records.

9.4 Baseline Anomaly Detection

Learn, then alert. The node observes a site for a set period and records what belongs there. Afterwards it reports what is new, what has gone, and what has changed signal.

```
@ALL BASELINE_START:300        # minimum 60s; add :FOREVER to run indefinitely
@ALL BASELINE_STATUS
```

Alerts: `ANOMALY-NEW`, `ANOMALY-RETURN`, `ANOMALY-RSSI`, `DEVICE_DISAPPEARED`.



Baseline in its learning phase: WiFi and BLE device counts, frames seen, and the running anomaly total.

9.5 MAC Randomization Correlation

Modern phones rotate their MAC address to defeat tracking. This mode links rotating MACs back to one persistent identity using behavioural signatures, and reports the identity rather than the address.

```
@ALL RANDOMIZATION_START:2:300
```

Emits `IDENTITY:` messages. Treat the output as an investigative lead, not proof. This is an experimental mode.

9.6 Deauth Attack Detection

Real-time detection of deauthentication and disassociation frames, with the source tracked.

```
@ALL DEAUTH_START:300
```

Alert: `NODE: ATTACK: DEAUTH|DISASSOC [BROADCAST|TARGETED] SRC:MAC DST:MAC RSSI:dBm CH:N R:reason`. The IEEE reason code is reported as context only: codes 1, 2, 6 and 7 all occur in normal traffic, so the firmware does not use the code to decide whether a frame is an attack.

9.7 Sentinel counter-intelligence

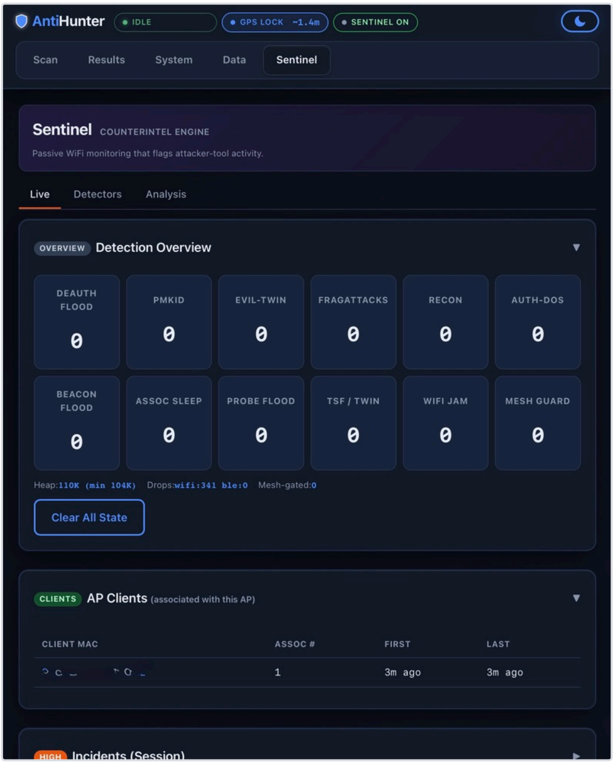
Passive detection of attacker *tooling*, not just attacks: deauth, beacon, auth and association floods, SAE denial of service, karma and evil-twin access points, probe floods, and handshake or PMKID capture attempts. Where a frame carries a recognisable template, Sentinel names the tool behind it.

```
@ALL SENTINEL_ON
@ALL SENTINEL_MODE:scan          # hop the configured channels
@ALL SENTINEL_BOOT:1             # start automatically on boot
@ALL GROUP:dos:on                # enable a detector group
@AH01 SENTINEL_STATUS
@AH01 INCIDENTS:50               # dump the incident log to serial
```

Group	Detectors
`dos`	eviltwin, sae, assoc_sleep
`rogue`	eviltwin, owe, karma
`recon`	pmkid, probe_flood, hshk
`physical`	frag, tsf, jam
`mesh`	mesh_guard
`all`	every member above

Deauth, beacon and auth detection are always on and have no toggle. `SENTINEL_MODE:defend` pins one channel. On Headless there is no AP to pin to, so use `scan`. Sentinel is WiFi only. BLE and drone detection are separate modes.

> **KARMA BAIT**
>
> Karma bait is the only part of the firmware that transmits: a probe request every 8 seconds to bait a karma AP into responding. It is off by default. Enable it deliberately with `GROUP:rogue:on` and only where transmitting is appropriate.



The Sentinel tab: a live counter per detector, the associated AP clients it has seen, and the session incident log.

9.8 Drone Remote ID

Identifies drones broadcasting Remote ID (ODID / ASTM F3411, and French ID) over both WiFi beacon/NAN and BLE. Reports the serial or CAA registration, position, altitude and speed, and the operator's location when the aircraft broadcasts it.

```
@ALL DRONE_START:300
```

`DRONE:` is sent once per appearance. A drone that stays in range is not re-announced. `DRONE_LOST:` follows 120 s after the last beacon.

9.9 Triangulation

Several nodes each report RSSI for one target MAC, and the coordinator fits a position from the set. Needs at least three nodes with valid GPS fixes and matched antennas.

```
@AH01 TRIANGULATE_START:AA:BB:CC:DD:EE:FF:60:2:1.0:1.0
@AH01 TRIANGULATE_RESULTS
@ALL TRIANGULATE_STOP
```

The environment argument sets the path-loss model: 0 open sky, 1 suburban, 2 indoor, 3 dense indoor, 4 industrial. Choosing the wrong one is the usual cause of a distance estimate that looks wrong. Results arrive as `T_D:` per-node data, `T_F:` the final fix with confidence and uncertainty, and `T_C:` complete, with a map link.

9.10 Reading the results

- **SD card, both builds.** Scan results in `/last_results.txt`, the probe database, vibration events in `/vibrations.jsonl`, sentinel incidents and the scan logs. Logging does not depend on which firmware you flashed.
- **Web UI, Full only.** The Results tab renders each scan as cards, and the API serves the same content as plain text.
- **Device database, Full only.** `/devicedb.jsonl` is what the Data Explorer browses. Headless does not write it.
- **Mesh.** Every alert above, plus `SCAN_DONE:` summaries carrying `TX=` broadcast and `DUP=` deduplicated counts.

> **EXPORTS CARRY LOCATION**
>
> SD logs, exports and mesh alerts can all contain GPS coordinates. Review any file before you post it publicly or attach it to a report. Privacy Mode only redacts the live web UI.

10 Vibration sensor and tamper response

10.1 How it works

The SW-420 is a mechanical vibration switch with a comparator and a trimpot on board. It has one job: pull its digital output high when movement exceeds the threshold the trimpot sets. That output goes to GPIO 2 on the ESP32, configured as a pull-down input with a rising-edge interrupt.

When it fires, the firmware:

- Sends a mesh alert if vibration alerts are enabled: `NODE: VIBRATION: Movement detected at TIME [GPS:lat,lon]`
- Writes a structured event to `/vibrations.jsonl` on the SD card
- Optionally starts a scan (§10.3)
- Counts toward the tamper auto-erase trigger if that is armed (§10.4)

Alerts are rate-limited to one every 3 seconds, so continuous shaking produces a steady trickle rather than a flood. The System diagnostics show `Vibration sensor: Standby` or `Vibration sensor: Active - Last: TIME (Ns ago)`.

Turn alerts on and off with `@AH01 VIBRATION_ON` and `VIBRATION_OFF`, or the toggle in the web UI's Sensor Alerts card. `VIBRATION_STATUS` reports the current state.

10.2 Tuning the trimpot

The trimpot is the small brass screw on the sensor module. It sets the sensitivity threshold, and the correct setting depends entirely on where the node is mounted. A pole in the wind is not a shelf indoors. Tune it **after** the node is in its final position, not on the bench.

1. Power the node and watch the sensor's own indicator LED, or open a serial monitor and watch for `[VIBRATION]` lines.
2. Leave the node completely still. If it is triggering by itself, turn the screw an eighth of a turn at a time in the direction that stops it. Wait several seconds between adjustments. The module reacts slowly.
3. When it is silent at rest, back off slightly further so ambient movement does not set it off.
4. Now tap the enclosure the way a person handling it would. It should fire.
5. If it does not, turn back the other way a fraction at a time until a deliberate tap triggers it but the node at rest does not.
6. Confirm the events actually reached the firmware: the diagnostics timestamp updates, and each event is appended to `/vibrations.jsonl`.

> **TUNING BY SYMPTOM**
>
> - **Constant false alerts.** Too sensitive, or mounted on something resonant. Reduce sensitivity, and consider a foam pad or a stiffer mount.
> - **Nothing fires even when you shake it.** Too insensitive, or the sensor is not seeing GPIO 2. Check the diagnostics line and the solder joints on the sensor header.
> - **Fires only sometimes.** You are on the edge of the threshold. Back it off a fraction toward sensitive.

10.3 Vibration auto-scan

Movement can start a scan automatically. This suits a node used as a trip sensor rather than a continuous monitor.

```
@AH01 VIBSCAN_SET:1:2:60:60    # enable : mode : duration s : cooldown s
@AH01 VIBSCAN_STATUS
```

Mode	Scan started	Mode	Scan started
0	None	4	List scan (targets)
1	All-device scan	5	Drone / Remote ID
2	Probe-request detect	6	Deauth (blue team)
3	Randomized-MAC detect	7	Baseline anomaly

Duration 0 means run until stopped. The trigger is skipped if a scan is already running or the node is in battery saver.

10.4 Tamper auto-erase

The node can destroy its own data when it is moved. This is **off by default**, and it is permanent and irreversible.

Parameter	Range	What it does
Setup delay	30 s to 10 min	Grace period after arming, so installing the node does not wipe it
Vibrations required	2 to 5, default 3	How many movement events count as tampering

Detection window	10 to 60 s	The period those events must fall within
Erase delay	10 to 300 s	Countdown before destruction, your window to cancel
Cooldown	5 to 60 min	Minimum time between tamper attempts, to blunt false triggers

```
@AH01 AUTOERASE_ENABLE:60:30:3:30:300   # setup:erase:vibs:window:cooldown
@AH01 AUTOERASE_STATUS
@AH01 AUTOERASE_DISABLE
@AH01 ERASE_CANCEL                       # abort a running countdown
```

During the setup period the node still reports movement, so you can watch the arming from a distance: `VIBRATION: Movement in setup mode (active in Ns)`. When it triggers you get `TAMPER_DETECTED: Auto-erase in Xs`.

Remote erase

Two steps, so a single stray message cannot wipe a node. Request a token, then use it. The token expires in five minutes and is specific to that device.

```
@AH01 ERASE_REQUEST
@AH01 ERASE_FORCE:AH_12345678_87654321_00001234
```

After a wipe the node plants a dummy IoT weather configuration in place of its own, so a recovered device does not obviously read as a sensor.

Factory reset

```
@AH01 FACTORY_RESET:FULL:yourEraseKey    # FULL | CONFIG | DATA
```

Single node only, and it requires the erase PSK you set in §8.1. Without a PSK set, there is no credential to authorise it.

> **DATA DESTRUCTION IS IRREVERSIBLE**
>
> Test your settings with a node that holds nothing you care about. A wipe cannot be undone, and the countdown is short by design.

11 Mesh command reference

Send these as ordinary text messages from the Meshtastic app, a Command Center, or any client on your channel. `@ALL` broadcasts. A node ID targets one node. All times UTC.

Core

Command	Description
STATUS	Mode, scan state, hits, temperature, uptime, GPS
STOP	Stop all operations, flush pending mesh transmissions

Configuration

Command	Parameters
CONFIG_TARGETS	Pipe-delimited MACs, OUI prefixes or SSIDs

CONFIG_NODEID	2 to 5 alphanumeric characters
CONFIG_RSSI	Threshold, −128 to −10
CONFIG_CHANNELS	Comma-separated list or range, e.g. 1..11
CONFIG_BAND	C5 only: 0 = 2.4 GHz, 1 = 5 GHz, 2 = both
CONFIG_DEDUP_TTL	0 to 3600 s. 0 disables cross-scan MAC dedup
CONFIG_SESSION_DEDUP	0 or 1, per-session dedup
MESH_DEDUP_CLEAR	Clear the dedup cache
CONFIG_ERASE_PSK	1 to 64 characters, authorises erase and factory reset

Scanning

Command	Parameters
SCAN_START	mode:secs:channels[:FOREVER], mode 0 WiFi, 1 BLE, 2 both
DEVICE_SCAN_START	mode:secs[:FOREVER[:+PROBE]]
PROBE_START and PROBE_STOP	mode:secs[:FOREVER][:+ALL]
BASELINE_START and BASELINE_STATUS	duration[:FOREVER], minimum 60 s
DRONE_START	secs[:FOREVER]
DEAUTH_START	secs[:FOREVER]
RANDOMIZATION_START	mode:secs[:FOREVER]
TRIANGULATE_START	target:duration[:rfEnv[:wifiPwr:blePwr]]
TRIANGULATE_RESULTS and TRIANGULATE_STOP	None

Sentinel

Command	Parameters
SENTINEL_ON, SENTINEL_OFF, SENTINEL_STATUS	None
SENTINEL_MODE	defend to pin a channel, or scan to hop. Persisted
SENTINEL_BOOT	1 or 0, auto-start on boot
GROUP	<dos|rogue|recon|physical|mesh|all>:<on|off>
DETECT_CFG and DETECT_CFG_GET	JSON tunables, 180 characters or fewer. GET dumps to serial
INCIDENTS and INCIDENTS_CLEAR	[:1–200], dump to serial or clear the log
ATTACKER_TRILAT	1 or 0, auto-triangulate a confirmed attacker's MAC. Off by default

Security, power and heartbeat

Command	Parameters
ERASE_REQUEST, ERASE_FORCE, ERASE_CANCEL	Token flow, §10.4

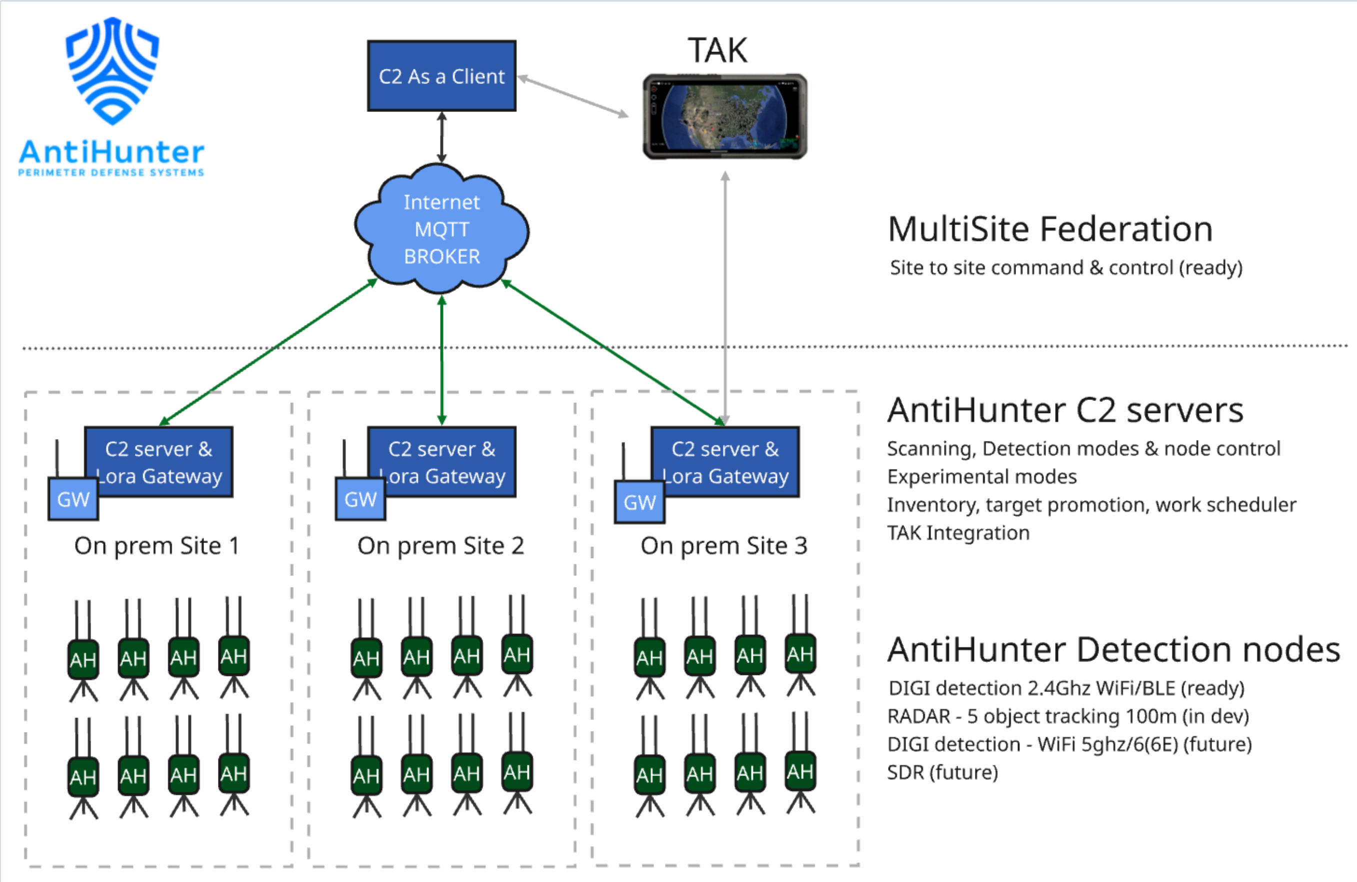
`AUTOERASE_ENABLE`	`setup:erase:vibs:window:cooldown`
`AUTOERASE_DISABLE` and `AUTOERASE_STATUS`	None
`VIBRATION_ON`, `VIBRATION_OFF`, `VIBRATION_STATUS`	None
`VIBSCAN_SET` and `VIBSCAN_STATUS`	`en:mode:dur[:cooldown]`
`FACTORY_RESET`	`<FULL|CONFIG|DATA>:<erase PSK>`
`BATTERY_SAVER_START`	Interval in minutes, 1 to 30, default 5
`BATTERY_SAVER_STOP` and `BATTERY_SAVER_STATUS`	None
`HB_ON`, `HB_OFF`, `HB_INTERVAL`	Periodic status broadcast, off by default. Interval 1 to 60 min

Ten commands to learn first

```
@ALL STATUS                          # who is alive, and what are they doing
@ALL STOP                            # stop everything, everywhere
@AH01 CONFIG_NODEID:GATE1            # name a node
@ALL CONFIG_TARGETS:AA:BB:CC|MyWiFi  # set the watchlist
@ALL SCAN_START:2:600:1..11          # 10 min watchlist scan, WiFi + BLE
@ALL DEVICE_SCAN_START:2:300         # survey everything nearby
@ALL BASELINE_START:900              # learn the site for 15 min
@ALL SENTINEL_ON                     # attacker-tool detection
@ALL DRONE_START:0                   # drone watch until stopped
@AH01 BATTERY_SAVER_START:10         # idle a node without losing the mesh
```

12 Command Center

One node needs no server. A fleet benefits from one: AntiHunter Command & Control PRO aggregates every node's detections onto a live map, with TAK, MQTT and ADS-B integration and a Sentinel console.



Nodes operate independently and coordinate over the mesh. The Command Center aggregates.

12.1 What you need

- A computer to run it on: Linux, macOS or Windows 10/11. It can be the same machine you flash from.
- A Meshtastic radio on USB, joined to **your** channel, to receive the mesh.
- Node.js 20 or newer and PostgreSQL. The installer puts both in place for you.

12.2 Quick install: Linux and macOS

```
curl -o setup-local.sh \
  https://raw.githubusercontent.com/TheRealSirHaXalot/AntiHunter-Command-Control-PRO/main/scripts/setup-local.sh
chmod +x setup-local.sh
./setup-local.sh
```

Run it as a normal user, **not** with sudo. It asks for sudo itself when it needs it. It installs the prerequisites, clones the project, sets up the database, runs the migrations and seeds an admin login. When it finishes, open **http://localhost:5173** and log in with the credentials it printed.

12.3 Quick install: Windows 10/11

1. Install Git in any PowerShell window, then close it so Git lands on your PATH:

   ```
   winget install -e --id Git.Git --accept-package-agreements --accept-source-agreements
   ```

2. Open PowerShell **as Administrator**. The installer needs it to install system software.
3. Clone and run the installer:

```
cd $HOME
git clone https://github.com/TheRealSirHaXalot/AntiHunter-Command-Control-PRO.git
cd AntiHunter-Command-Control-PRO
powershell -ExecutionPolicy Bypass -File scripts\setup-windows.ps1
```

4. Press Enter to accept the defaults. The first run takes a few minutes. **Write down the admin email and password it shows you.**
5. Start it with **Start-AntiHunter.cmd** in the project folder, or `pnpm AHCC`.
6. Open **http://localhost:5173** and log in.

> **WINDOWS NOTES**
>
> - The firewall prompt on first start: click **Cancel** if you only use it on this PC, since localhost works either way. To reach it from other devices, click Allow and tick **Private** only.
> - For serial hardware, put `SERIAL_DEVICE=COM3`, your port from Device Manager → Ports, in `apps\backend\.env` and restart.
> - Do not use WSL. COM ports do not pass through cleanly.

12.4 Docker

Needs Docker 25 or newer. If you have no serial adapter at `/dev/ttyUSB0`, which is every macOS and Windows host, comment out the `devices:` and `group_add:` lines under the `backend` service first, or the container will not start.

```
git clone https://github.com/TheRealSirHaXalot/AntiHunter-Command-Control-PRO.git
cd AntiHunter-Command-Control-PRO
docker compose up -d --build
```

Three containers come up: Postgres, the backend on port 3000, and the frontend on **http://localhost:8080**. The seeded admin defaults to `admin@example.com` and `admin`. Logs: `docker compose logs -f backend`. Stop: `docker compose down`, adding `--volumes` to delete the database too.

> **CHANGE THE ADMIN PASSWORD IMMEDIATELY**
>
> The seeded credentials are published in the documentation. Change them at first login, before the server is reachable by anything but you. Enable two-factor authentication and HTTPS if the instance will be reached across a network, and work through the hardening checklist in the project README.

12.5 Updating

Use the in-app update page, or `git pull` in the project folder. Docker: `git pull && docker compose up -d --build`. The backend re-applies migrations on boot.

13 Deployment

13.1 Siting

- **Height helps more than power.** Get the node above metal, vehicles and crowds. A few metres of elevation beats any antenna change.
- **Antennas vertical**, and clear of the enclosure's own metalwork and of each other.

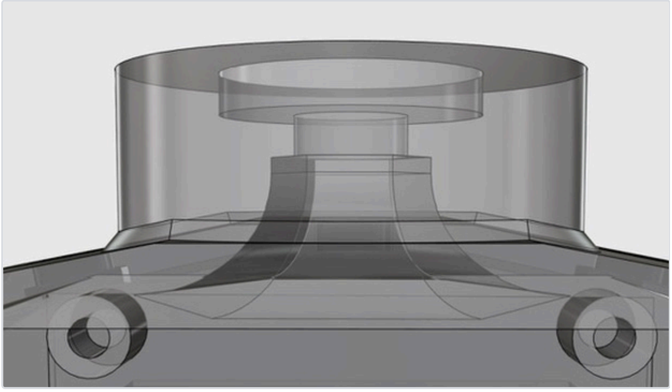
- **GPS wants sky.** The helix or patch needs a clear upward view. Indoors it may never lock, and without a fix your detections carry no coordinates and triangulation cannot run.
- **Keep LoRa and WiFi antennas apart** as far as the enclosure allows.
- For multi-node triangulation, spread nodes around the area rather than in a line, and give each one a clear GPS fix. Matched antennas on every node.

Mast, tripod, can and puck builds. The enclosure changes. The node does not.

13.2 Sealing and mounting

- Fit the TPU seals for the housing, GPS and USB-C port. Silicone grease on the seals and SMA threads helps in persistent wet.
- No IP rating is claimed for the printed enclosure. Treat it as weather-resistant, not waterproof, and site it accordingly.
- The base takes a ¼"-20 insert for tripods and the universal bracket. STL files for brackets, solar mounts and radar extensions are in the `hw/` folder.
- Point the vented lid and fan opening downward or shielded so water does not sit in them.

An enclosed build with the antennas vertical and the switch centred.



Vented lid, SMA bulkheads on top, battery housing below.

13.3 Power planning

- Roughly 160 mA idle, 600 mA scanning. Aggressive presets and continuous promiscuous capture sit at the top of that range.
- Battery Saver stops scanning, drops the CPU to 80 MHz and polls GPS once a minute while keeping the mesh alive. That is the right state for a node that only needs to be reachable.
- Solar mount frames are in the STL folder. Feed solar through the UPS board, never directly to the PCB.
- Never use the chassis USB input and the UPS fast-charge input at the same time.

13.4 Before you walk away

- ☐ AP credentials changed, or Headless firmware flashed
- ☐ LoRa region set and your own encrypted channel primary, public channel off
- ☐ Node ID set and recorded
- ☐ Clock set, GPS has a fix
- ☐ SD card mounted and writable
- ☐ Vibration sensor tuned in its final position
- ☐ Auto-erase parameters understood if you armed it, and the erase PSK recorded somewhere safe
- ☐ The node answers `@NODEID STATUS` over the mesh from where you will be standing
- ☐ Antennas tight, seals seated, lid closed

14 Maintenance and troubleshooting

14.1 Symptom table

Symptom	Likely cause	What to do
No `Antihunter` WiFi network	Headless firmware flashed, or the AP name and password were changed	Check over serial which build is running. Reflash Full if you need the UI.
Web UI will not load	Not connected to the node's own AP, or using https	Join the node's AP, then open **http://192.168.4.1** exactly.
Flashing fails or the port never appears	Charge-only USB cable, a serial monitor still holding the port, both USB ports connected, or the wrong browser	Use a data cable, close other serial tools, unplug the radio, use Chrome or Edge.

SD card not detected	Not FAT32, a 32 GB or larger card, or not seated	Reformat FAT32 on a card under 32 GB. Reseating means opening the enclosure.
No GPS fix	Antenna not connected, or no view of the sky	Check the U.FL or SMA connection, move outdoors, give it time. The module runs hot while searching, which is normal.
Nothing arrives over the mesh	Region `UNSET` so the radio is receive-only, serial module off, wrong pins for the board, or nodes on different channels or keys	Re-run `meshtastic_config.py` and print the settings. Confirm every node carries the same channel.
Node hears commands but never answers	Region unset on *that* node, so it cannot transmit	Set its region.
Constant vibration alerts	Trimpot too sensitive, or a resonant mount	Retune in place (§10.2), pad the mount.
Vibration never fires	Trimpot too insensitive, or a bad joint on the sensor header	Retune, check the diagnostics line and the solder joints.
Node scans forever	A scan was started with `:FOREVER` or duration 0	`@ALL STOP`, or the Stop button in the UI.
Settings gone after flashing	Erase Device was ticked	Reconfigure. Settings mirror to the SD card and self-heal, but a deliberate erase clears NVS.
Node hot to the touch	Normal under sustained scanning	Fit the fan and thermal switch, ventilate the enclosure, consider a relaxed preset.
Detections with no coordinates	No valid GPS fix at the time	Expected behaviour. Position is only attached when the fix is valid.

14.2 Routine maintenance

- Pull the SD card periodically and archive the logs, then clear them, so a lost node carries less.
- Check the cells. A swollen or damaged 18650 is retired immediately, never recharged.
- Re-check antenna tightness and seal condition after any period in the field.
- Reflash on the same channel to pick up fixes. Read the release notes first.

14.3 Getting help

Issues and Discussions at **github.com/lukeswitz/AntiHunter**. A useful report includes: firmware build and channel, Full or Headless, board type, what you did, what you expected, what happened, and the serial output around the failure. Scrub coordinates and MAC addresses from anything you paste. See §15.

15 Legal and safety

15.1 Authorisation

AntiHunter is distributed for lawful, authorised defensive use only: security research, blue-team training, compliant monitoring, event security and network auditing. Operate it only on infrastructure, networks, devices, spectrum and datasets you own or have explicit written permission to assess. The detection, correlation and triangulation features exist to characterise an environment you are authorised to assess, not to identify, follow or profile people. Targeted surveillance, stalking, harassment and tracking individuals without informed consent are prohibited.

15.2 Radio compliance

You are responsible for every rule governing radio use where you operate: FCC Part 15 and Part 97, CE/RED, Ofcom and their national equivalents; ISM band allocations, power limits and duty cycles; and any licensing conditions for your class of operation. LoRa is region-locked, 868 MHz EU, 915 MHz US, 923 MHz Asia, and running a region outside the one you are in may be unlawful. Never use the node to cause harmful interference.

15.3 Interception and privacy

Passive reception of radio emissions is regulated separately from network access in most jurisdictions. Capturing, storing, decoding or disclosing frame contents and identifiers may engage the US Wiretap Act and ECPA, state two-party-consent statutes, the UK Investigatory Powers Act or equivalents. Determine the lawfulness of each capture mode where you are before enabling it.

MAC addresses, probe request contents, BLE advertisements and Remote ID broadcasts may be personal data under GDPR, UK GDPR, CCPA/CPRA and similar regimes. Collect only with a lawful basis, minimise what you keep, set a retention and destruction schedule, and honour data-subject rights. The maintainers never receive, process or host your data.

15.4 Drone Remote ID

Receiving broadcast Remote ID is permitted in most places. Using that data to locate, approach, confront or interfere with an aircraft or its operator may violate aviation, harassment or anti-stalking law, including 18 U.S.C. § 32. Reception is not authorisation to act.

15.5 Hardware and battery safety

Kits, bare boards and assembled units are supplied for people competent in electronics assembly and operation. Correct assembly, ESD control, antenna selection and attachment, and supply of regulated 5 V power are your responsibility. Operating the transceiver without a properly matched antenna can damage the hardware. Lithium cells are never supplied. Sourcing, protection, charging, storage, transport and disposal are yours, and carry fire and injury risk. The hardware is not certified for and must not be used in life-safety, medical, aviation, automotive or industrial-control applications. It is not a substitute for a certified security or alarm system, and its output is advisory only, subject to false positives and false negatives.

15.6 Experimental features

Modes marked beta or experimental, including Sentinel, Triangulation and MAC Randomization Correlation, are unvalidated, may produce inaccurate or misleading output, and must not be relied on for operational, evidentiary, investigative or safety decisions.

DISCLAIMER

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Assembly, sale or instruction by the maintainers creates no warranty, no certification, no fitness-for-purpose representation and no responsibility for how a unit is later configured, deployed or used. Once the unit is in your hands, operation and compliance are yours alone. Flashing the firmware, by web flasher, shell script or any other means, is acceptance of these terms.

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> You alone are responsible for ensuring your build, deployment and operation comply with all applicable laws, regulations, licences, permits, equipment authorisations, organisational policies and third-party rights. **If you do not agree to these terms, do not build, flash, assemble, deploy or operate AntiHunter.**
>
> Full text: `Legal Disclaimer` in the project README. Firmware is licensed under the GNU AGPL v3.0. Hardware and documentation are licensed as stated in their own files.

Appendix A · Pinout

XIAO ESP32-S3 on the DIGI NODE board. Pin assignments may change with board revisions. Verify against your revision.

Function	GPIO	Notes
Vibration sensor	2	SW-420 tamper detection, rising-edge interrupt, input pull-down
RTC SDA	3	DS3231 I²C data
RTC SCL	6	DS3231 I²C clock
GPS RX	44	NMEA in
GPS TX	43	Unused
SD CS	1	Chip select
SD SCK	7	SPI clock
SD MISO	8	SPI MISO
SD MOSI	9	SPI MOSI
Mesh RX	4	From the Meshtastic radio
Mesh TX	5	To the Meshtastic radio

Radio side: serial module in TEXTMSG mode at 115200 baud, pins 19 RX / 20 TX on Heltec V3, 10 RX / 9 TX on T114.

Appendix B · Bill of materials

Core

- 1× AntiHunter DIGI PCB, 82 mm, 2-layer
- 1× Seeed XIAO ESP32-S3, 8 MB flash minimum, or XIAO ESP32-C5 for 2.4 and 5 GHz
- 1× Heltec WiFi LoRa 32 V3.2, T114 also compatible
- 1× ATGM336H GPS module
- 1× DS3231 RTC module
- 1× SW-420 vibration sensor
- 1× micro SD reader module
- 1× SD card, 8 GB supplied on every tier but the Bare PCB. FAT32, 32 GB and larger not recommended
- 1× KSD9700 normally-open thermal switch, 40 °C

Connectors and fasteners

- 6× JST 2.54 2-pin terminals

- 3× JST power male cables for the switch, thermal switch and power board
- 10× M3 heat-set inserts
- 4× M2 heat-set inserts for the power board
- 2× M3 15 mm brass standoffs
- M3×4-6mm screws for the PCB and enclosure lids, M2×4-6mm for the power board
- 2× M2.5 screws, supplied with the fan
- 1× ¼"-20 tripod insert

Antenna and cabling

- 3× U.FL to SMA pigtail, 10 to 20 cm
- SMA bulkheads
- 1× 6 dBi 2.4 GHz antenna
- 1× 6 dBi LoRa antenna for your region
- 1× GPS antenna. Passive ceramic patch on the kit and populated PCB, active helix on the assembled unit

Power and thermal

- 1× 30 mm 5 V fan with JST lead
- 1× 3-pin mini on/off switch
- 1× waterproof Type-C female chassis jack, 2-pin 22 AWG leads, 14 mm panel nut, dust cap
- 1× Type-C 15 W 3 A 5 V UPS board, dual 18650
- 1× 2 A fast-blow fuse, recommended, not supplied
- 2× 18650 cells, protected flat-top, never supplied with any tier

Enclosure

- 3D-printed weatherproof enclosure. STLs, seals, brackets and solar mounts in `hw/Prototype_STL_Files/`

Part links and photographs: `hw/Prototype_STL_Files/BOM-Links.md`

Appendix C · Defaults, addresses and files

Item	Default	
WiFi AP name	`Antihunter`	**change it**
WiFi AP password	`antihunt3r123`	**change it**
Web UI	`http://192.168.4.1`	Full firmware only
Erase PSK	not set	**set it** if using erase or factory reset
Meshtastic serial	TEXTMSG, 115200	pre-configured on tiers 1 and 3, you set it on 2 and 4
LoRa region	`UNSET`, receive only	**set it**
Meshtastic channel	public default	**replace with your own encrypted channel**
Scan preset	Balanced	
WiFi channels	1 to 11	
Mesh dedup TTL	300 s	0 disables
Vibration alerts	enabled	rate-limited to one per 3 s
Vibrations required for tamper	3	range 2 to 5

Auto-erase	disabled	
Heartbeat	disabled	
Karma bait	off	the only transmitting detector
Attacker auto-triangulation	off	
C2 frontend	`http://localhost:5173`, Docker 8080	
C2 seeded admin	`admin@example.com` and `admin`	**change at first login**

Files on the SD card

Path	Contents
`/last_results.txt`	Most recent scan results as text. Written by both builds
`/devicedb.jsonl`	Device database. Full firmware only, browsed by the Data Explorer
`/vibrations.jsonl`	Structured vibration events

Appendix D · Glossary

BSSID	The MAC address of an access point.
OUI	The first three bytes of a MAC address, identifying the manufacturer. Watchlisting an OUI matches a whole vendor.
Probe request	A frame a device sends looking for networks it knows. Reveals remembered SSIDs.
Ghost SSID	A probed network with no matching AP responding nearby.
Evil twin	A rogue AP impersonating a legitimate network to capture clients.
Karma	An AP that answers every probe request, claiming to be whatever a client asks for.
Deauth	A management frame that disconnects a client. Used to force reconnections and capture handshakes.
PMKID	A value in the first EAPOL message that can be captured and cracked offline without a full handshake.
Remote ID	The broadcast identification a drone emits under ASTM F3411 and equivalent rules.
RSSI	Received signal strength, in dBm. Less negative is stronger.
NVS	Non-volatile storage on the ESP32 where settings persist across reboots.
SoftAP	The WiFi access point the Full firmware hosts for its web UI.
U.FL / IPX	The miniature snap-on RF connector on each module.

Appendix E · Quick reference

Print this page and keep it with the node.

Never

- Power from anything but a regulated 5 V source
- Power the radio with no LoRa antenna fitted
- Connect both USB ports at once
- Press a U.FL connector at an angle
- Leave the default AP password or the public mesh channel in place

Addresses

Web UI	`http://192.168.4.1`, AP `Antihunter`
Web flasher	`lukeswitz.github.io/AntiHunter`
Firmware, docs, issues	`github.com/lukeswitz/AntiHunter`
Command Center	`github.com/TheRealSirHaXalot/AntiHunter-Command-Control-PRO`
Meshtastic flasher	`flasher.meshtastic.org`

Ten commands

```
@ALL STATUS                         who is alive
@ALL STOP                           stop everything
@AH01 CONFIG_NODEID:GATE1           name a node
@ALL CONFIG_TARGETS:AA:BB:CC|MyWiFi set the watchlist
@ALL SCAN_START:2:600:1..11         watchlist scan, 10 min
@ALL DEVICE_SCAN_START:2:300        survey everything
@ALL BASELINE_START:900             learn the site
@ALL SENTINEL_ON                    attacker-tool detection
@ALL DRONE_START:0                  drone watch until stopped
@AH01 BATTERY_SAVER_START:10        idle, mesh stays up
```

Setup order for a new node

1. Fit the antennas
2. Flash AntiHunter, Full to start with
3. Change the AP credentials
4. Set the radio region
5. Create your encrypted channel and disable the public one
6. Set the node ID and the clock
7. Confirm the SD card and GPS
8. Tune the vibration trimpot in place
9. Verify `@NODEID STATUS` answers over the mesh
10. Seal and deploy